

Занулити підвідрізок

Назва вхідного файлу:	standard input
Назва вихідного файлу:	standard output
Ліміт часу:	6 seconds
Ліміт використання пам'яті:	512 megabytes

Це задача з градерами.

Для масиву додатних цілих чисел b довжини m визначимо $f(b)$ наступним чином:

- нехай початково деяка змінна x рівна 0;
- за одну *монету* дозволяється збільшити значення x на 1;
- за одну *монету* дозволяється обрати елемент масиву b_i ($1 \leq i \leq m$) та замінити його на $(b_i \oplus x)$, де $\oplus$ позначає операцію *бітового виключного АБО*;
- $f(b)$ дорівнює мінімальній кількості монет, необхідній для того, щоб зробити всі елементи масиву b одночасно рівними нулю.

Бітове виключне АБО невід'ємних цілих чисел a та b ($a \oplus b$) дорівнює невід'ємному цілому числу, у якого у двійковому записі на певній позиції знаходиться одиниця тоді і тільки тоді, коли у двійкових записах a та b на цій позиції знаходяться різні значення. Наприклад, $3_{10} \oplus 5_{10} = 0011_2 \oplus 0101_2 = 0110_2 = 6_{10}$.

Задано масив додатних цілих чисел a довжини n та q запитів виду l, r . Для кожного запиту потрібно знайти $f([a_l, a_{l+1}, \dots, a_r])$.

Протокол взаємодії

Вам потрібно реалізувати наступні функції:

`void init(integer n, array of integers a)`

- n — ціле число, яке позначає довжину масиву;
- a — масив цілих чисел довжиною n ;
- ця функція нічого не повертає.

`integer ask(integer l, integer r)`

- l — ціле число, яке позначає ліву межу запиту;
- r — ціле число, яке позначає праву межу запиту;
- ця функція повертає ціле число — $f([a_l, a_{l+1}, \dots, a_r])$.

`array of integers askAll(integer q, array of integers l, array of integers r)`

- q — ціле число, яке позначає кількість запитів;
- l — масив цілих чисел довжиною q ; l_i позначає ліву межу i -го запиту;
- r — масив цілих чисел довжиною q ; r_i позначає праву межу i -го запиту;
- ця функція повертає масив цілих чисел; i -те число має бути рівне відповіді на i -й запит.

Формат вхідних даних

У першому рядку задано три цілі числа n , q та t ($1 \leq n, q \leq 2 \cdot 10^5$; $1 \leq t \leq 2$) — кількість чисел, кількість запитів та формат запитів відповідно.

У другому рядку задано n цілих чисел $a_1, a_2, \dots, a_n$ ($1 \leq a_i < 2^{60}$) — елементи масиву a .

У наступних q рядках задано по два цілі числа l та r ($1 \leq l \leq r \leq n$) — параметри i -го запиту.

Спочатку буде викликана функція `init` рівно один раз.

Якщо $t = 1$, то буде викликана рівно один раз функція `askAll` з усіма запитами. Якщо $t = 2$, то функція `ask` буде викликана рівно q разів.

Формат вихідних даних

Градер виведе у окремих рядках q цілих чисел — відповіді на запити.

Система оцінювання

- (3 бали): $t = 1$, $a_i = a_1$ для $1 \leq i \leq n$;
- (8 балів): $t = 1$, $a_i \neq a_j$ для $i \neq j$;
- (3 бали): $t = 1$, $2^m + n \leq a_i < 2^{m+1}$ для деякого натурального m ;
- (9 балів): $t = 1$, $a_i \leq a_{i+1}$ для $1 \leq i < n$;
- (10 балів): $t = 1$, $n, q \leq 1000$;
- (11 балів): $t = 1$, $l_i = 1$ та $r_i = i$ для $1 \leq i \leq q$.
- (10 балів): $t = 1$, $n, q \leq 50000$;
- (25 балів): $t = 1$;
- (9 балів): $t = 2$, $n, q \leq 10^5$;
- (12 балів): $t = 2$.

Приклади

standard input	standard output
7 6 1 5 4 3 5 7 7 7 1 4 4 7 3 7 1 7 2 6 1 1	9 11 12 14 12 6
7 6 2 5 4 3 5 7 7 7 1 4 4 7 3 7 1 7 2 6 1 1	9 11 12 14 12 6

Problem B. Increment and Swap

Time Limit 2000 ms

Problem Statement

We have a sequence A of length N .

On this sequence, we can perform the following two kinds of operations:

- Swap two adjacent elements.
- Select one element, and increment it by 1.

We will repeatedly perform these operations so that A will be a non-decreasing sequence. Find the minimum required number of operations.

Constraints

- $1 \leq N \leq 200000$
- $1 \leq A_i \leq 10^9$
- A_i is an integer.

Input

Input is given from Standard Input in the following format:

```
N
A1
A2
:
AN
```

Output

Print the minimum number of operations required to turn A into a non-decreasing sequence.

Sample 1

Input	Output
5 4 1 8 8 7	2

We can turn A into a non-decreasing sequence in two operations:

- Initially, $A = \{4, 1, 8, 8, 7\}$.
- Swap the first two elements. Now, $A = \{1, 4, 8, 8, 7\}$.
- Increment the last element by 1. Now, $A = \{1, 4, 8, 8, 8\}$.

Sample 2

Input	Output
20 8 2 9 7 4 6 7 9 7 4 7 4 4 3 6 2 3 4 4 9	62

Problem C

Potential Peak

Time limit: 6 seconds

A running club is recruiting members and growing rapidly, with exactly one new member joining each day. Every member comes with a specific initial skill level, and they are assigned a unique ID based on the order of their arrival.

At the end of every day, the Coach likes to calculate a metric he calls the “total potential” of the club. This value is simply the sum of the individual potentials of all current members. To keep the members motivated, the potential of a member is calculated through a hypothetical “what-if” scenario involving a single, intense workout.

In this imaginary workout, a member can choose to put in a certain amount of effort to temporarily boost their skill level. For every unit of effort they expend, their skill level increases by exactly one unit. In return, they gain one unit of “happiness” for every other member currently in the club who:

- joined the club strictly after they did; and
- was originally strictly more skilled than they were but has now been caught or surpassed thanks to the workout.

Note that during this calculation, only that member’s skill level is increased, while the skill levels of all other members remain fixed at their original values.

The potential of a single member is defined as the maximum possible value of their net gain (happiness minus effort) they could achieve by choosing their effort level optimally. These workouts are entirely theoretical; they are used only for the Coach’s daily report, and the members’ actual skill levels remain unchanged from day to day.

Your task is to help the Coach calculate the total potential of the club at the end of each day of the recruiting period.

Input

The first line of input contains an integer n ($1 \leq n \leq 10^5$), representing the number of days the club is recruiting members.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$), where a_i is the initial skill level of the member who joins on day i .

Output

Output n integers in a single line. The k -th integer should be the total potential of the club at the end of day k .

Sample Input 1

8
1 2 3 1 2 3 2 3

Sample Output 1

0 0 0 0 1 3 5 9

Explanation of Sample 1: On day 5, there are 5 members with skill levels $[1, 2, 3, 1, 2]$. For member 1: There are 3 members joining later with strictly higher skills: member 2, member 3, and member 5. The maximum potential for member 1 is 1, achieved by either of the following strategies:

- If member 1 pays 1 unit of effort, their skill becomes 2. They will catch or surpass member 2 and member 5. The happiness is 2 units, and the gain is $2 - 1 = 1$ unit.
- If member 1 pays 2 units of effort, their skill becomes 3. They will catch or surpass members 2, 3, and 5. The happiness is 3 units, and the gain is $3 - 2 = 1$ unit.

The other members have maximum potential 0. Hence, the total potential for day 5 is 1.

On day 8, there are 8 members with skill levels $[1, 2, 3, 1, 2, 3, 2, 3]$. The optimal efforts are $[2, 1, 0, 2, 1, 0, 1, 0]$ and the corresponding maximum potentials are $[4, 2, 0, 2, 1, 0, 0, 0]$.

Sample Input 2

1
1

Sample Output 2

0

Problem D. 比赛

Time Limit 2000 ms

Mem Limit 524288 kB

Description

有 n 名选手，编号分别为 $1, 2, \dots, n$ 。编号为 i 的选手的实力值为 i 。所有选手分为红、蓝两队，其中编号为 i 的选手所在的队伍用字符 s_i 描述。 s_i 为 **R** 表示他在红队， s_i 为 **B** 表示他在蓝队。

现在将所有选手排成一个环，每一对相邻且不属于同一队的选手会进行一场比赛，实力值较大的选手获胜，他所在的队伍的得分增加一。

然而，蓝队的选手勾结了裁判，如果一场比赛中红队选手获胜，且他在蓝队选手的顺时针方向上，则这场比赛不计入得分。

现在你想知道，对于每个 $k = -n, \dots, -1, 0, 1, \dots, n$ ，有多少种将选手排列的方法，使得红队得分恰好比蓝队得分大 k 。两种方案不同，当且仅当存在某个选手，使得他顺时针方向的下一个选手在两个方案中不同。

由于答案很大，你只需要输出答案对 $998,244,353$ 取模后的值。

Input

从标准输入读入数据。

第一行包含一个整数 n ，表示选手个数。

第二行包含一个长度为 n 的字符串 $s_1 s_2 \dots s_n$ ，表示每个选手所属的队伍。

Output

输出到标准输出。

输出一行 $2n + 1$ 个整数，分别表示 $k = -n, \dots, 0, \dots, n$ 的方案数对 $998,244,353$ 取模后的值。

Sample 1

Input	Output
3 BRB	0 0 1 1 0 0 0

Sample 2

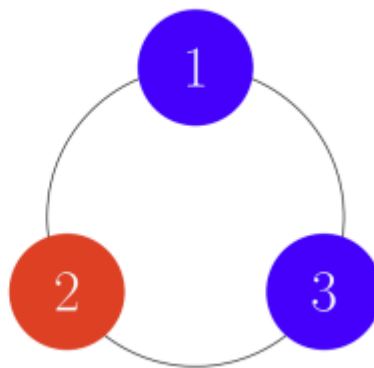
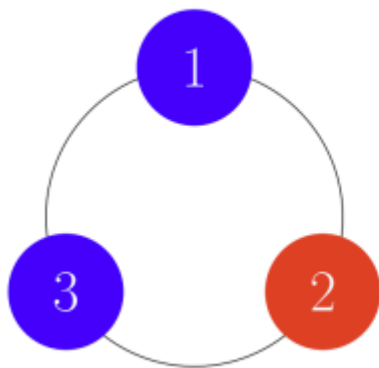
Input	Output
5 RBBRR	0 0 0 0 8 8 8 0 0 0 0

Sample 3

Input	Output
见题目目录下的 3.in 与 3.ans。	见题目目录下的 3.in 与 3.ans。

Hint

【样例 1 解释】



洛谷

如图所示，共有两种排列的方法。

第一种排列中，选手 1 与 2 之间的比赛虽然选手 2 获胜，但他属于红队，且在选手 1 顺时针方向，故这场比赛不计入得分。而选手 2 和 3 之间的比赛蓝方获胜。因此红队得分为 0，蓝队得分为 1。

第二种排列中，共进行两场比赛，且均计入得分。红队得分与蓝队得分均为 1。

【子任务】

对于所有数据，满足 $3 \leq n \leq 3000$ ， s 为由 **B** 和 **R** 构成的字符串。

子任务编号	分值	n
1	10	≤ 17
2	10	≤ 30
3	10	≤ 50
4	10	≤ 200
5	45	≤ 500
6	15	≤ 3000

Problem E. 乘积的期望

Time Limit 2500 ms

Mem Limit 1024000 kB

Description

有一个长度为 n 的序列 $a_1, a_2, \dots, a_n$ 。初始序列的所有元素均为 0。再给定正整数 m 、 c 和 $(n - m + 1)$ 个正整数 $b_1, b_2, \dots, b_{n-m+1}$ 。

对序列 $a_1, a_2, \dots, a_n$ 进行 c 次操作，每次操作为：

- 随机选择整数 $1 \leq x \leq n - m + 1$ ，其中选到 y ($1 \leq y \leq n - m + 1$) 的概率为 $\frac{b_y}{\sum_{i=1}^{n-m+1} b_i}$ 。将 $a_x, a_{x+1}, \dots, a_{x+m-1}$ 增加 1。

c 次操作中对 x 的随机是独立的。

求操作完成后序列中所有元素的乘积的期望。为了避免浮点数输出，你需要将答案对 998244353 取模。

Input

从标准输入读入数据。

第一行三个整数 n, m, c ，分别表示序列长度、操作区间长度和操作次数。

第二行 $(n - m + 1)$ 个整数 $b_1, \dots, b_{n-m+1}$ ，描述随机的权重。

Output

输出到标准输出。

输出一行一个整数，表示 c 次操作后序列中所有数的乘积的期望。

Sample 1

Input	Output
3 2 2 1 1	1

Sample 2

Input	Output
10 3 10 1 2 3 4 5 6 7 8	721023399

Sample 3

Input	Output
20 12 98765 9 8 7 6 5 4 3 2 1	560770686

Hint**【样例 1 解释】**

当两次操作选择的 x 不同时，最终序列为 $[1, 2, 1]$ ，序列元素乘积为 2；否则序列为 $[2, 2, 0]$ 或 $[0, 2, 2]$ ，序列元素乘积均为 0。两次操作选择的 x 不同的概率为 $\frac{1}{2}$ ，因此输出 $2 \times \frac{1}{2} = 1$ 。

子任务

对于所有测试数据， $2 \leq m \leq n \leq 50$ ， $1 \leq c < 998244353$ ，对于 $1 \leq i \leq n - m + 1$ ， $1 \leq b_i \leq 10^5$ 。

- Subtask 1 (10%) : $m \leq 8$ 。
- Subtask 2 (20%) : $m \leq 16$ 。
- Subtask 3 (15%) : $n \leq 20, c \leq n$ 。
- Subtask 4 (15%) : $n \leq 30, c \leq n$ 。
- Subtask 5 (15%) : $n \leq 40, c \leq n$ 。
- Subtask 6 (15%) : $c \leq n$ 。
- Subtask 7 (10%) : 无特殊限制。

Problem F. 能量分配

Time Limit 4000 ms

Mem Limit 1048576 kB

Background

来自 2026 清华大学学生程序设计竞赛暨高校邀请赛 (THUPC2026) 初赛。

题解等资源可在 <https://gitlink.org.cn/thusaa/thupc2026pre> 查看。

Description

小 Q 是星际小学光速飞船课的老师，她准备向学校的星际能源商店购买 n 个**本质不同**的能量石，其中每个能量石具有一单位能量，且每个能量石有一个 1 到 n 之间的正整数编号，每个能量石的编号互不相同。她想要把它们分给 m 名学生，帮助学生们练习如何驾驶光速飞船。

在分配时，每个能量石会恰好被分给一名学生，且允许学生**没有分配**到任何能量石。定义 c_i 表示第 i 名学生分配到的能量石总数，即满足 $\sum_{i=1}^m c_i = n$ 且 $c_i \geq 0$ 。两个能量石的分配方案本质不同当且仅当**存在一个学生在两个方案中分配到的能量石的编号集合不同**。例如对于两个能量石与两名学生，能量石 1 分给学生 1，能量石 2 分给学生 2 与能量石 1 分给学生 2，能量石 2 分给学生 1 被视为两种本质不同的方案，由于学生 1 在方案一中分配到的能量石的编号集合为 $\{1\}$ 而在方案二中分配到的能量石的编号集合为 $\{2\}$ ，所以被视为本质不同的方案。

然而，每名学生都想要自己分配到的用于光速飞船的能量石比其他学生分到的多，因此如果一个学生分配到的能量石比其他学生分到的能量石要更多，那么他会将拥有更大的满意度。

初始时每名学生的满意度为 1，则在分配完 n 个能量石后第 i 名学生的满意度可以按照如下方式计算：

- 依次考虑每一个除了 i 自己以外的学生 j ，若 $c_i > c_j$ ，则 i 的满意度将会变为原来的 $A_{i,j}$ 倍；若 $c_i = c_j$ ，则 i 的满意度将会变为原来的 $B_{i,j}$ 倍；若 $c_i < c_j$ ，则 i 的满意度将不会发生变化。保证满足 $1 < B_{i,j} < A_{i,j} \leq 10^9$ 。

对于第 i 名学生，令其在分配完 n 个能量石后的满意度为按照如上方式考虑完除自己以外的 $m - 1$ 名学生后最终的满意度。现在小 Q 定义一个能量石的分配方案的权值为在分完 n 个能量石后 m 名学生**满意度之积**，小 Q 想知道所有本质不同的能量分配方案的**权值和对 317 取模的结果**。然而小 Q 是光速飞船课老师，不是光速计算课老师，所以她希望精通光速计算的您，迅速告诉她这个结果对 317 取模的结果。

此外，由于小 Q 事先没有测试星际小学光速飞船的具体性能，当光速飞船的性能不同时，能量需求会发生变化。所以她会询问 q 个可能购买的能量石总数 n ，对于每一个 n ，您需要告诉她所有若她购买了 n

个能量石，所有能量石的分配方案的权值和对 317 取模的结果。

Input

从标准输入读入数据。

第一行输入一个正整数 $m(1 \leq m \leq 12)$ ，表示学生总数。

接下来的 m 行输入 $m \times m$ 的矩阵 A ，其中每行 m 个正整数，第 i 行第 j 列表示 $A_{i,j}$ 的取值，其中当 $i = j$ 时 $A_{i,j} = 0$ ，当 $i \neq j$ 时有 $1 < A_{i,j} \leq 10^9$ 。

接下来的 m 行输入 $m \times m$ 的矩阵 B ，其中每行 m 个正整数，第 i 行第 j 列表示 $B_{i,j}$ 的取值，其中当 $i = j$ 时 $B_{i,j} = 0$ ，当 $i \neq j$ 时有 $1 < B_{i,j} < A_{i,j}$ 。

之后一行输入一个正整数 $q(1 \leq q \leq 100)$ ，表示 q 个询问。

接下来的 q 行每行输入一个正整数 $n(1 \leq n \leq 10^{18})$ ，表示询问当能量石总数为 n 时原问题的答案。

Output

输出到标准输出。

输出 q 行，每行一个 $[0, 317)$ 之间的整数，表示能量石的分配方案权值和对 317 取模的结果。

Sample 1

Input	Output
3 0 7 5 5 0 7 4 6 0 0 4 4 3 0 4 3 2 0 4 1 2 10 10000000000000000000	37 303 46 148

Hint

样例一解释

对于 $n = 1$ ，共有 3 种本质不同的能量石的分配方案，即将唯一的能量石分配给学生 1, 2 或 3，每种方案的权值分别为 280, 420, 288，所有分配方案的权值和为 988，对 317 取模的结果为 37。

对于 $n = 2$ ，共有 9 种本质不同的能量石的分配方案，对于每一组满足 $1 \leq i, j \leq 3$ 的正整数对 (i, j) ，能量石 1 分配给学生 i ，能量石 2 分配给学生 j 对应一组方案，各方案的权值如下表，其中第 i 行第 j 列的数表示将能量石 1 分配给学生 i ，能量石 2 分配给学生 j 所对应的分配方案的权值：

280	420	504
420	420	160
504	160	288

各方案的权值和为 3156，对 317 取模的结果为 303。

Problem G. Arkady and Rectangles

Time Limit 4000 ms

Mem Limit 262144 kB

Arkady has got an infinite plane painted in color 0. Then he draws n rectangles filled with paint with sides parallel to the Cartesian coordinate axes, one after another. The color of the i -th rectangle is i (rectangles are enumerated from 1 to n in the order he draws them). It is possible that new rectangles cover some of the previous ones completely or partially.

Count the number of different colors on the plane after Arkady draws all the rectangles.

Input

The first line contains a single integer n ($1 \leq n \leq 100\,000$) — the number of rectangles.

The i -th of the next n lines contains 4 integers x_1, y_1, x_2 and y_2 ($-10^9 \leq x_1 < x_2 \leq 10^9$, $-10^9 \leq y_1 < y_2 \leq 10^9$) — the coordinates of corners of the i -th rectangle.

Output

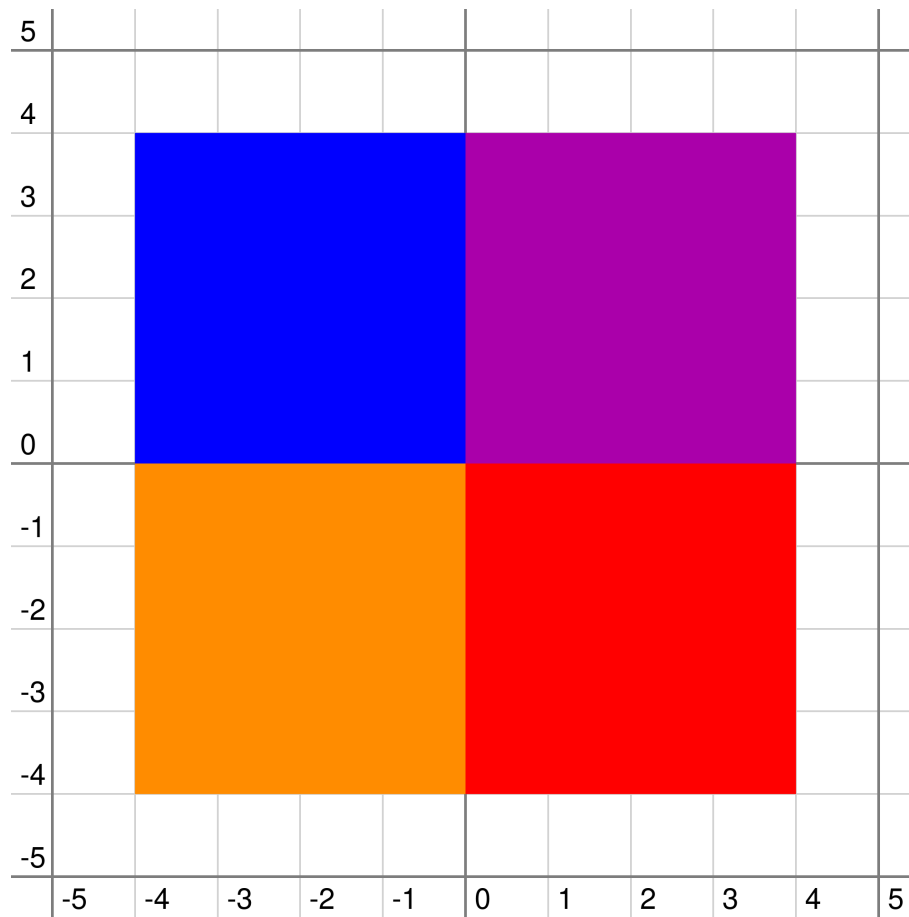
In the single line print the number of different colors in the plane, including color 0.

Examples

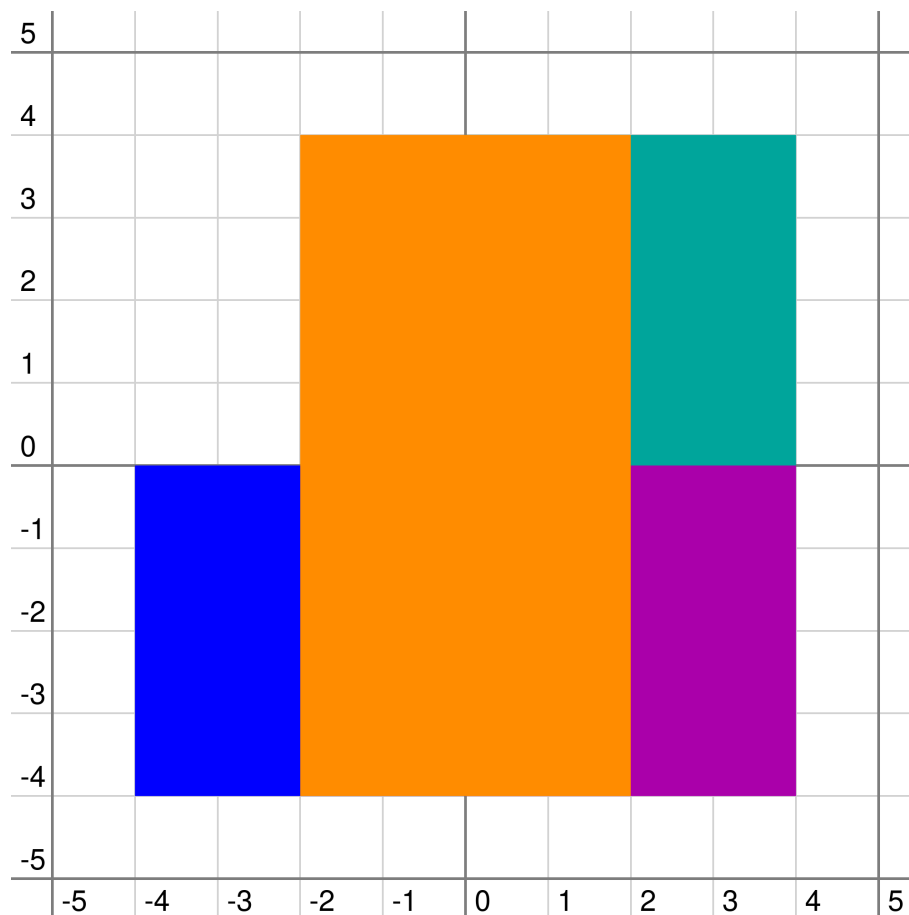
Input	Output
<pre>5 -1 -1 1 1 -4 0 0 4 0 0 4 4 -4 -4 0 0 0 -4 4 0</pre>	5

Input	Output
<pre>4 0 0 4 4 -4 -4 0 0 0 -4 4 0 -2 -4 2 4</pre>	5

Note



That's how the plane looks in the first sample



That's how the plane looks in the second sample

0 = white, 1 = cyan, 2 = blue, 3 = purple, 4 = yellow, 5 = red.

Problem H. 健身房 / Siłownia

Time Limit 8100 ms

Mem Limit 524288 kB

Background

译自 [PA 2015](#) R5.

Description

健身房里有 k 个器材。

有 n 个人预约了健身。第 i 个人预约给定了 l_i, r_i, p_i ，意思是要分配给他 $l_i, l_i + 1, \dots, r_i$ 中的一个（记为 x ），他在第 x 个小时中用器材 p_i 健身。

同一时间不能有两个人用同一个健身器材。此外，老板还希望让健身房里没人的时刻尽量多，这样可以节约电费。

构造一组最优解。

Input

第一行，两个正整数 n, k 。

接下来 n 行，第 i 行三个正整数 l_i, r_i, p_i 。

Output

如果无解，输出一行一个 **NIE**。

否则输出 $(n + 1)$ 行：第一行一个整数，表示在最优策略下，有多少个时刻健身房至少有一个人。接下来 n 行，第 i 行的整数表示第 i 个人分配的 x 。

Sample 1

Input	Output
4 2 1 3 1 1 1 1 1 3 2 3 3 2	2 3 1 1 3

Sample 2

Input	Output
3 1 1 2 1 1 2 1 1 2 1	NIE

Hint

- $1 \leq n \leq 10^6$;
- $1 \leq k \leq 10^9$;
- $1 \leq l_i \leq r_i \leq 10^9$;
- $1 \leq p_i \leq k$.

Alice, Bob, And Two Arrays

Input file: standard input
Output file: standard output
Time limit: 2.5 seconds
Memory limit: 1024 megabytes

There is an array a of length N and an array b of length M . All numbers in the arrays are integers and lie in the range from 1 to k . There is also an initially empty array c .

Alice and Bob play the following game on these arrays: the players take turns, and on their turn, a player must append a number to the end of the array c such that c remains a subsequence of both a and b . The player who cannot make a move loses. Alice goes first.

They will play the game q times. For the i -th game, they will choose two numbers x_i and y_i ($0 \leq x_i < N, 0 \leq y_i < M$), then remove the first x_i elements from array a and the first y_i elements from array b , and then play the game on the resulting arrays. After each deletion operation and before the next one, the arrays a and b are restored to their initial state, meaning that the numbers deleted from the arrays in one game may not be deleted in subsequent games. Also, the array c is cleared between games.

The guys have their quirks, so they always choose x_i and y_i in such a way that after the deletions, **the remaining parts of arrays a and b start with the same value**.

Alice really wants to win, so she asks you to determine for each game whether she can win, assuming both players play optimally.

Note that the arrays can be very long, so they are provided in a special format. Each array is given as a sequence of segments of equal numbers. Array a consists of n such segments, and array b consists of m such segments. Each segment is defined by its length and the number that occupies that segment.

Input

The first line contains six integers N, n, M, m, k, q ($1 \leq N, M \leq 10^9, 1 \leq n, m, k \leq 1600, 1 \leq q \leq 10^6$) — the length of the first array, the number of segments in the first array, the length of the second array, the number of segments in the second array, the number limit, and the number of games, respectively.

The next n lines contain two integers l_i^a and v_i^a ($1 \leq l_i^a \leq N, 1 \leq v_i^a \leq k$) — the length of the segment and the number written in that segment. These numbers define array a : the first l_1^a numbers of array a are equal to v_1^a , the next l_2^a numbers are equal to v_2^a , ..., the last l_n^a numbers are equal to v_n^a .

The next m lines contain two integers l_i^b and v_i^b ($1 \leq l_i^b \leq M, 1 \leq v_i^b \leq k$) — the length of the segment and the number written in that segment. These numbers define array b . The format is similar to array a .

It is guaranteed that $\sum l_i^a = N, \sum l_i^b = M, v_i^a \neq v_{i+1}^a$, and $v_i^b \neq v_{i+1}^b$.

The next q lines contain pairs of integers x_i and y_i ($0 \leq x_i < N, 0 \leq y_i < M$) — descriptions of the games.

For each game i , it is guaranteed that if we remove the first x_i elements from a and the first y_i elements from b , the remaining parts of the arrays will start with the same value.

Output

For each of the q games, print “Yes“ if Alice wins with optimal strategy, and “No“ if Bob wins.

Examples

standard input	standard output
5 1 5 1 1 9	Yes
5 1	No
5 1	Yes
0 0	No
0 1	No
0 2	Yes
1 0	Yes
1 1	Yes
1 2	Yes
2 0	
2 1	
2 2	
7 3 7 3 2 12	Yes
2 1	No
3 2	Yes
2 1	Yes
2 2	No
3 1	Yes
2 2	No
0 2	Yes
0 3	No
0 4	Yes
1 2	Yes
1 3	Yes
1 4	
2 5	
2 6	
3 5	
3 6	
4 5	
4 6	

Note

In the first example, the arrays look like this: $a = (1, 1, 1, 1, 1)$ and $b = (1, 1, 1, 1, 1)$.

- In the first query, $x = 0, y = 0$, so the game will be played on the arrays $a = (1, 1, 1, 1, 1)$ and $b = (1, 1, 1, 1, 1)$. In this case, players can only append the number 1 to the array c , so after 5 moves the game will end, and Bob will lose because he won't be able to make a move.
- In the second query, $x = 0, y = 1$, so the game will be played on the arrays $a = (1, 1, 1, 1, 1)$ and $b = (1, 1, 1, 1)$. In this case, the game will end after 4 moves, and Alice will lose.
- In the last query, $x = 2, y = 2$, so the game will be played on the arrays $a = (1, 1, 1)$ and $b = (1, 1, 1)$. In this case, Bob will lose.

In the second example, $a = (1, 1, 2, 2, 2, 1, 1)$, $b = (2, 2, 1, 1, 1, 2, 2)$.

- In the first query, $x = 0$ and $y = 2$, so the game will be played on the arrays $a = (1, 1, 2, 2, 2, 1, 1)$ and $b = (1, 1, 1, 2, 2)$. If Alice appends the number 2 to the array c , Bob will also append the number 2, and then no moves will be left, so Alice will lose. Therefore, Alice must first append the number 1

to c . After this, for similar reasons, if Bob appends the number 2 to the array a , he will lose. So, he is forced to append the number 1, and the array $c = (1, 1)$. Then Alice again appends the number 1 to the array c , and Bob has no more moves left, so Alice wins.

- In the second query, $x = 0$ and $y = 3$, so the game will be played on the arrays $a = (1, 1, 2, 2, 2, 1, 1)$ and $b = (1, 1, 2, 2)$. Following the reasoning in the previous example, Alice cannot append the number 2 to the array c , because then she will lose. But if Alice appends the number 1, then Bob will also append the number 1, and after that, Alice will lose for similar reasons. Therefore, Bob wins in this case.

Scoring

The tests for this problem consist of eleven groups. Points for each group are given only if all tests of the group and all tests of the required groups are passed. Please note that passing the example tests is not required for some groups. **Offline-evaluation** means that the results of testing your solution on this group will only be available after the end of the competition.

Group	Points	Additional constraints			Required Groups	Comment
		N, M	n, m	q		
0	0	–	–	–	–	Examples.
1	13	$N, M \leq 300$	–	$q \leq 10^5$	0	
2	12	$N, M \leq 5000$	–	$q \leq 10^5$	0, 1	
3	11	–	–	$q \leq 10^5$	–	$l_i^a \leq 1000$ and all v_i^a are distinct $l_i^b \leq 1000$ and all v_i^b are distinct
4	8	–	–	$q \leq 10^5$	3	$l_i^a \leq 1000$ and all v_i^a are distinct
5	10	–	–	$q \leq 10^5$	–	$l_1^a \geq N - 500$ and $v_1^a = 1$ $l_1^b \geq M - 500$ and $v_1^b = 1$
6	7	$N, M \leq 10^5$	$n, m \leq 100$	$q \leq 10^5$	–	$k \leq 5$
7	6	$N, M \leq 10^5$	$n, m \leq 100$	$q \leq 10^5$	0, 6	$k \leq 50$
8	7	–	$n, m \leq 100$	$q \leq 10^5$	0, 6, 7	$k \leq 50$
9	9	–	$n, m \leq 800$	$q \leq 10^5$	0, 6 – 8	
10	10	–	–	$q \leq 10^5$	0 – 9	Offline-evaluation.
11	7	–	–	–	0 – 10	Offline-evaluation.

Problem G. Guess sinus

Input file: `standard input`
Output file: `standard output`
Time limit: `1 second`
Memory limit: `256 megabytes`

This is an interactive problem. Jury choose a real number a ($|a| \leq 10^9$) with no more than 5 digits after decimal point and you need to guess it. You could send jury queries of two types:

1. Send real number x , jury will send you $\text{sign}(\sin(a \cdot x))$.
2. Send real number y trying to guess number a .

You could send not more than 200 queries of first type and only one query of second type. Your answer will be considered correct if its absolute or relative error will not exceed 10^{-6} .

Input

For each your query of first type a single line will be printed, containing one number: 0, 1 or -1. There will be no response for second type query.

Output

In each line print one your query.

Print "`? x`" to get $\text{sign}(\sin(a \cdot x))$ or print "`! y`" if you think that answers is y .

You may use no more than 200 queries of the first type.

Your program should terminate immediately after it sends a query of the second type.

Examples

standard input	standard output
1	? 1
-1	? -1
	! 1

Note

$\text{sign}(\sin(a \cdot x))$ will be calculated using standard C++ functions from `math.h` and `double` datatype.

Among Us

Input file: **standard input**
Output file: **standard output**
Time limit: **1 second**
Memory limit: **256 megabytes**

Aboard the spaceship *Skeld* are you — the ship's commander — along with N crewmates. The voyage from Earth to *Polus* was meant to be peaceful... until it wasn't. A series of strange incidents occurs: oxygen runs low, the reactor nearly melts down, the lights flicker, and the doors jam shut. Suspicion spreads — there might be impostors among us.

Moments of crisis were handled with the crewmates' unity, but the tension lingered. Each crewmate now secretly suspects exactly **one** crewmate of being the impostor. Remarkably, **no crewmate is suspected by more than one person**. Formally, there exists a permutation P of length N , where $P[i]$ is the crewmate whom the i -th crewmate suspects.

Overseeing this chaos from the commander's room, you, however, are not actually concerned with finding the impostors. Rather, **your only goal is to determine the hidden permutation P** .

Naturally, to achieve this, you may call **emergency meetings**. In each meeting, you can choose the order in which the crewmates speak.

Given a crewmate speaking, they will accuse the crewmate they suspect. If the speaker has not yet been marked as **sus**, their accusation is valid, and the accused crewmate becomes **sus**. However, if the speaker is already **sus**, their accusation is ignored.

Unfortunately, as the commander, you cannot attend the meetings yourself. Instead, after each meeting, the ship's log reports the *sus* status of all crewmates as a binary array, where 1 indicates **sus** and 0 indicates **not sus**.

Since time is limited, you may hold at most 400 emergency meetings. Your task is to determine the hidden permutation P within this limit.

Interaction Protocol

Your program first reads a single integer N ($2 \leq N \leq 100$) — the number of crewmates.

Then, you may repeatedly perform queries (call emergency meetings). To make a query, print a line in the following format:

- ? $q_1 q_2 \dots q_N$

Here, $(q_1, q_2, \dots, q_N)$ is the order in which the crewmates will speak during the meeting. The sequence must be a permutation of length N .

After printing a query, flush the output and read a line containing N integers:

- $s_1 s_2 \dots s_N$

Here, s_i indicates the status of the i -th crewmate after the meeting ends:

- $s_i = 0$ — the i -th crewmate is not **sus**.
- $s_i = 1$ — the i -th crewmate is **sus**.

Each emergency meeting is independent; all crewmates start as not sus.

When you have determined the hidden permutation P , print it in the following format:

- ! $p_1 p_2 \dots p_N$

Printing the answer does **not** count as a query. After printing the final answer, your program should immediately terminate. Failure to do so may result in an undefined verdict.

You may perform at most **400 queries**. If your program performs more than 400 queries or outputs an invalid format, you may receive a **Wrong Answer** verdict.

The interactor is **not adaptive**, which means that the hidden permutation does not depend on the queries you make.

After printing a query or an answer, do not forget to output the end of line and flush the output. Otherwise, you may get an **Idleness Limit Exceeded** verdict. To do this, use:

- `fflush(stdout)` or `cout.flush()` in C/C++;
- `System.out.flush()` in Java and Kotlin;
- `sys.stdout.flush()` in Python;

Example

standard input	standard output
6	? 1 2 3 4 5 6
1 0 1 1 1 0	? 2 4 3 1 5 6
0 0 1 1 1 1	? 6 5 4 3 2 1
1 1 1 0 0 1	? 3 5 4 2 1 6
0 1 1 1 0 1	! 4 5 3 6 2 1

Note

For the first emergency meeting, the permutation P is $[4, 5, 3, 6, 2, 1]$ and the accusations go as follows:

- Crewmate 1 accuses crewmate 4. Crewmate 4 becomes *sus*.
- Crewmate 2 accuses crewmate 5. Crewmate 5 becomes *sus*.
- Crewmate 3 accuses crewmate 3. Crewmate 3 becomes *sus*.
- Crewmate 4 accuses crewmate 6, but they are already *sus*, so it is ignored.
- Crewmate 5 accuses crewmate 2, but they are already *sus*, so it is ignored.
- Crewmate 6 accuses crewmate 1. Crewmate 1 becomes *sus*.
- The ship reports the status of the crewmates as 1 0 1 1 1 0.

Problem J. Just Another Disney Problem (14+)

Evil wizard Jafar has a huge collection of lamps. He likes touching them, wiping the dust, looking at his reflection in them. He loves all of them almost equally, they are very beautiful, but for every two of them he likes one of them more than another one. Jafar keeps his lamps in a very long hall, all lamps in one line. One day he decided to arrange all lamps along his way from one side of the hall to another in such a way, that for every two neighboring lamps he likes the next one more than previous. In other words Jafar would like a some kind of an ascending order of lamp's quality. You are a new servant of wizard and you should fulfill the desire of your master. The main problem is that you don't know anything about Jafar's preferences. You can ask Jafar for any two lamps which one is better, but you should be careful, he is very busy now in his plans for world domination and you shouldn't ask him too much questions. Note that preferences could be non-transitive. You should output desired order of all lamps or report Jafar that it doesn't exist.

Input

First number — N ($1 \leq N \leq 1000$). Answer for every question — string “YES”, if Y is better than X , and “NO”, if X is better than Y .

Output

Your questions — one line with three integers $1, X, Y$ ($1 \leq X, Y \leq N, X \neq Y$). You should ask not more than 10 000 questions. In the last line: integer 0, then N integers a_i ($1 \leq a_i \leq N$) — the desired permutation or N zeros if such permutation doesn't exist. All integers in the lines should be separated by spaces.

Examples

stdin	stdout
3	1 1 2
YES	1 2 3
NO	1 1 3
NO	0 3 1 2

Note

The pipe from your program to the interactor program and the pipe back have limited size. Your program must read from the standard input to avoid deadlock. Deadlock condition is reported as Wall time-limit exceeded.

To flush the standard output stream use the following statements:

In C use `fflush(stdout);`

In C++ use `cout.flush();`

In Java use `System.out.flush();`

If your program receives EOF (end-of-file) condition on the standard input, it MUST exit immediately with exit code 0. Failure to comply with this requirement may result in Time-limit exceeded error.

Problem E. Guess Table

Input file: *standard input*
Output file: *standard output*
Time limit: 2 seconds
Memory limit: 256 mebibytes

Jury has created a $n \times m$ matrix A consisting of zeroes and ones. You should guess it.

In order to do that you can send no more than 200 queries. At the end you should print the desired matrix.

Each query is described by the matrix B of size $a \times b$ ($1 \leq a \leq n, 1 \leq b \leq m$) consisting of zeroes, ones and questions. The answer to the query is 1 if the matrix A contains continuous submatrix that corresponds to the given pattern B . Otherwise the answer will be equal to 0.

Formally the answer will be equal to 1 if and only if exists some values (i, j) ($1 \leq i \leq n - a + 1, 1 \leq j \leq m - b + 1$), so for each (x, y) ($1 \leq x \leq a, 1 \leq y \leq b$) B_{xy} is a question or $B_{xy} = A_{i+x-1, j+y-1}$.

Also there is another restriction for the query matrix B : there should be no row and no column fully consisting of questions.

Note the problem is interactive, so you should flush your output after each query and at the end of the guess. To do that use function `fflush(stdout)` in C++ language and `PrintWriter.flush()` in Java language.

Input

You are given two integers n and m ($1 \leq n, m \leq 13$) — the size of the matrix to guess.

Output

At the end you should print a line with a string **"Answer:"** (without quotes) on the first line. After that print n lines with m symbols "0" and "1" — the desired matrix. Don't forget to flush you output after that.

Interaction Protocol

On the first line of each your query print a string **"Query:"** (without quotes) and two integers a, b ($1 \leq a \leq n, 1 \leq b \leq m$) — the size of the submatrix. After that print a lines with b symbols "0", "1" and "?". Don't forget to flush you output after that.

After each query you will get an answer — a single line with number 1 if the desired matrix contains given submatrix and 0 otherwise.

Example

standard input	standard output
2 3	Query: 2 1 0 0
1	Query: 1 2 01
1	Query: 2 2 01 0?
1	Query: 2 3 101 ??0
0	Answer: 101 001

Kevin and Teams

Input file: **standard input**
Output file: **standard output**
Time limit: 4 seconds
Memory limit: 256 megabytes

This is an interactive problem.

Kevin has n classmates, numbered $1, 2, \dots, n$. Any two of them may either be friends or not friends.

Kevin wants to select $2k$ classmates to form k teams, where each team contains exactly 2 people. Each person can belong to at most one team.

Let u_i and v_i be two people in the i -th team. To avoid potential conflicts during team formation, the team members must satisfy one of the following two conditions:

- For all i ($1 \leq i \leq k$), classmate u_i and v_i are friends.
- For all i ($1 \leq i \leq k$), classmate u_i and v_i are not friends.

Kevin wants to determine the maximum k such that, regardless of the friendship relationships among the n people, he can always find $2k$ people to form the teams. After that, he needs to form k teams. But asking whether two classmates are friends is awkward, so Kevin wants to achieve this while asking about the friendship status of no more than n pairs of classmates.

The interactor is **adaptive**. It means that the hidden relationship between classmates is not fixed before the interaction and will change during the interaction.

Interaction Protocol

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains one positive integer n ($2 \leq n \leq 10^5$) — the number of classmates.

First, you should output an integer k ($1 \leq k \leq \frac{n}{2}$) — the maximum number of teams you can form.

You can make queries in the following way — print one line of the form `? u v` where $1 \leq u \neq v \leq n$. After that, read a single integer: 0 or 1 indicating whether they are friends. 1 means they are friends while 0 means not.

If you want to print the answer, output `! u1 v1 u2 v2 ... uk vk`. You should output **exactly** $2k$ distinct numbers. Then, the interaction continues with the next test case.

You can make at most n queries. Printing the answer does **not** count towards the number of queries made.

It is guaranteed that the sum of n over all test cases does not exceed 10^5 .

After printing each query do not forget to output the end of line and flush* the output. Otherwise, you will get **Idleness limit exceeded** verdict.

If, at any interaction step, you read `-1` instead of valid data, your solution must exit immediately. This means that your solution will receive **Wrong answer** because of an invalid query or any other mistake. Failing to exit can result in an arbitrary verdict because your solution will continue to read from a closed stream.

Hacks

*To flush, use:

- `fflush(stdout)` or `cout.flush()` in C++;
- `sys.stdout.flush()` in Python;
- see the documentation for other languages.

The interactor for hacks is not adaptive. To make a hack, use the following format.

The first line contains the word “`manual`”.

The second line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains an integer n ($2 \leq n \leq 10^5$) — the number of classmates.

The second line of each test case contains a string s consisting of ‘0’ and ‘1’ of length $\frac{n(n-1)}{2}$ — indicating the relationship between $(1, 2), (1, 3), \dots, (1, n), (2, 3), (2, 4), \dots, (2, n), \dots, (n - 1, n)$. ‘1’ means being friends and ‘0’ means not being friends.

The following is the input of example.

```

manual
2
3
011
5
1011101011

```

The sum of n over all test cases must not exceed 10^5 .

There is an additional constraint for hacks: The sum of $\frac{n(n-1)}{2}$ over all test cases must not exceed 10^7 .

Example

standard input	standard output
2	
3	
	1
	? 1 2
1	
	! 1 2
5	
	2
	? 1 2
1	
	? 3 4
0	
	? 3 5
1	
	? 1 3
0	
	? 2 4
0	
	! 1 2 3 5

Note

In the first test case:

Kevin claims he can form 1 team regardless of the friendship relationships among the 3 people.

Kevin asks about the friendship relationship between people 1 and 2. The jury responds that they are friends.

Kevin answers that he can form a team with people 1 and 2.

In the second test case:

Kevin claims he can form 2 teams regardless of the friendship relationships among the 5 people.

Kevin asks about the friendship relationship between people $(1, 2)$, $(3, 4)$, $(3, 5)$, $(1, 3)$, $(2, 4)$. The jury responds with 1, 0, 1, 0, 0.

Kevin answers that he can form two teams with people $(1, 2)$ and $(3, 5)$.

It is also possible to form two teams with people $(1, 3)$ and $(2, 4)$, since they are both not friends.

Problem O. Library game

Time Limit 2000 ms

Mem Limit 2097152 kB

Description

Alessia and Bernardo are discovering the world of competitive programming through the books of their university library.

The library consists of m sections numbered from 1 to m . Each section contains only books dedicated to a particular subject and different sections correspond to different subjects. In order to prevent the students from wandering in the library, the university has established a system of passes. Each pass has a length y associated to it and allows access to an interval of y consecutive sections in the library. During a visit, the student must choose exactly one book from one of these sections and leave the library. Each pass can be used only once.

At the moment Alessia and Bernardo have n passes of lengths $x_1, x_2, \dots, x_n$. They have different opinions on the best way to improve: Alessia thinks that it is important to study many different topics, while Bernardo believes that it is important to study deeply at least one topic. So, Alessia wants to use the n passes to get n books on distinct topics, while Bernardo would like to get at least two books on the same topic.

They have reached the following agreement: for each of the following n days, Alessia will choose a pass of length y among those which are still available and an interval of y sections in the library, and Bernardo will go into the library and will take exactly one book from one of those sections.

Can Bernardo manage to get at least two books on the same subject, or will Alessia be able to avoid it?

You should decide whether you want to be Alessia or Bernardo, and you have to fulfill the goal of your chosen character. The judge will impersonate the other character. Note that, even if at some moment Bernardo has already taken two books on the same subject, the interaction should go on until the end of the n days.

Input

The first line contains two integers n and m ($1 \leq n \leq 100, n \leq m \leq 5000$) — the number of passes and the number of sections.

The second line contains n integers $x_1, x_2, \dots, x_n$ ($1 \leq x_i \leq m$) — the lengths of the passes available.

Interaction

First, you should print a line containing either the string **Alessia** or the string **Bernardo** — the character that you want to impersonate.

Then, for each of the n turns:

- If you are taking the role of Alessia, print a single line consisting of two integers y and a ($1 \leq a \leq m - y + 1$) — you are choosing a pass of length y and the interval of sections $[a, a + y - 1]$. Note that at least one pass of length y must still be available. After this, read a single integer b ($a \leq b \leq a + y - 1$) — the subject of the book selected by Bernardo.
- If you are taking the role of Bernardo, read two integers y and a ($1 \leq a \leq m - y + 1$) — the length of the pass chosen by Alessia and the index of the first section of the interval. It is guaranteed that there is at least one pass of length y available. Then, print a line consisting of a single integer b ($a \leq b \leq a + y - 1$) — the subject from which you choose to take a book.

If one of your interactions is malformed, the interactor terminates immediately and your program receives the verdict **WRONG-ANSWER**. Otherwise, you will receive the verdict according to the game's criteria described above.

After printing a line do not forget to end the line and flush the output. Otherwise, you will get the verdict **TIMELIMIT**. To flush the output, use:

- `fflush(stdout)` in C;
- `fflush(stdout), cout << flush` or `cout.flush()` in C++;
- `System.out.flush()` in Java and Kotlin;
- `sys.stdout.flush()` in Python.

Sample 1

Input	Output
5 14 3 7 2 3 10	-

Sample 2

Input	Output
4 10 4 1 6 4	-

Hint

In the first sample, it can be shown that Alessia can accomplish her goal. An example of interaction (after reading the input) follows:

Contestant	Judge	Explanation
Alessia		The program will act as Alessia
3 11		Choose $y = 3$ and $a = 11$
	13	Judge selects $b = 13$
10 2		Choose $y = 10$ and $a = 2$
	9	Judge selects $b = 9$
7 1		Choose $y = 7$ and $a = 1$
	4	Judge selects $b = 4$
2 10		Choose $y = 2$ and $a = 10$
	10	Judge selects $b = 10$
3 6		Choose $y = 3$ and $a = 6$
	7	Judge selects $b = 7$

The program of the contestant wins because all the books chosen by Bernardo pertain to different topics. The actions performed by the contestant and the judge in this example of interaction may be non-optimal.

In the **second sample**, it can be shown that Bernardo can manage to fulfil his goal. An example of interaction (after reading the input) follows:

Contestant	Judge	Explanation
Bernardo		The program will act as Bernardo
	4 1	Judge chooses $y = 4$ and $a = 1$
4		Select $b = 4$
	1 10	Judge chooses $y = 1$ and $a = 10$
10		Select $b = 10$
	6 3	Judge chooses $y = 6$ and $a = 3$
4		Select $b = 4$
	4 5	Judge chooses $y = 4$ and $a = 5$
8		Select $b = 8$

The program of the contestant wins because Bernardo has selected two books on topic number 4. The actions performed by the contestant and the judge in this example of interaction may be non-optimal.

Problem P. 神经衰弱 / Memory2

Time Limit 2000 ms

Mem Limit 262144 kB

Background

翻译参考自 [LibreOJ](#)。

Description

有 $2N$ 张卡片，每张卡片正面写有一个 0 到 $N - 1$ 之间的整数，且每个整数恰好出现两次。你和 JOI 君正在使用这 $2N$ 张卡片进行名为“神经衰弱”的游戏练习。

游戏练习开始时，所有卡片背面朝上，横向排成一行放在桌面上。从左往右第 $i + 1$ 张卡片 ($0 \leq i \leq 2N - 1$) 称为卡片 i ，其正面所写的整数记为 A_i ($0 \leq i \leq 2N - 1$)。最初，你和 JOI 君都不知道 A_i ($0 \leq i \leq 2N - 1$) 的具体数值。

你和 JOI 君最多可以重复进行 K 轮如下操作：

1. 你指定 $2N$ 张卡片中的任意两张卡片。
2. JOI 君将这两张卡片翻面，但确保你无法看到卡片正面所写的整数。如果两张卡片正面所写的整数相等，JOI 君会记住这个值并告诉你；否则，他会从两张卡片正面所写的整数中，选择一个对他而言更容易记住的值告诉你。

JOI 君对整数的记忆难度由 N 个整数 $P_0, P_1, \dots, P_{N-1}$ 表示，这些整数满足以下两个条件：

- $0 \leq P_i \leq N - 1$ ($0 \leq i \leq N - 1$)。
- $P_i \neq P_j$ ($0 \leq i < j \leq N - 1$)。

对 JOI 君而言，整数 i 比整数 j 更容易记住，当且仅当 $P_i < P_j$ 成立。

你的任务是：通过与 JOI 君进行最多 K 轮操作，确定每张卡片正面所写的整数。但你并不知道 JOI 君用于表示记忆难度的整数 $P_0, P_1, \dots, P_{N-1}$ 的具体取值。

题目

请编写一个程序，通过与 JOI 君进行交互操作，确定每张卡片正面所写的整数。

实现细节

你需要写一个程序实现确定每张牌上写的数字的功能。你不需要引入外部头文件，但是你必须声明函数 `int Flip(int I, int J);` 和 `void Answer(int I, int J, int X);`。并且，你应当选择

不低于 C++17 的语言标准提交。

程序中必需实现以下函数：

- `void Solve(int T, int N)`

对于每组测试数据，这个函数仅调用一次。 T 代表子任务编号， N 代表有 $2N$ 张牌。

这个函数必需通过调用 `Flip` 函数来确定牌上写的数字，通过调用 `Answer` 函数来做出回答。

程序中可以调用以下函数：

- `int Flip(int I, int J)`

当指定 JOI 君翻哪两张牌时调用这个函数。参数 I, J 表示 JOI 君需要翻开纸牌 I 和 J 。

I 和 J 都必须是 0 到 $2N - 1$ （包括两端）的整数，并且 I 不等于 J 。如果调用 `Flip` 函数时不满足条件，则会被判为 **Wrong answer [1]**。

如果 $A_I = A_J$ ，则这个函数返回这个值，否则会返回 A_I 和 A_J 中更容易记住的那个值。

如果这个函数调用超过 K 次，则会被判为 **Wrong answer [2]**。

- `void Answer(int I, int J, int X)`

这个函数表示写有整数 X 的卡片可以被确定了。

参数 I, J, X 需要满足以下条件：

- $0 \leq I, J \leq 2N - 1$
- $I \neq J$
- $A_I = A_J = X$

如果调用时参数不满足以上条件，则会被判为 **Wrong answer [3]**。

调用时，参数 X 的值需要与先前任意调用中的 X 不同。如果不满足，则会被判为 **Wrong answer [4]**。

此函数必须恰好被调用 N 次，否则会被判为 **Wrong answer [5]**。

你的程序可以实现任何其他函数，或定义全局变量。但你的程序无论如何都不可以与标准输入输出或其他文件交互。

附加文件中包含一个样例交互器和交互库，仅用作测试。

样例交互器包含一个文件，文件名为 `grader.c` 或 `grader.cpp`。为了测试程序，需执行如下命令：

- C 语言

```
1 | gcc -std=c11 -O2 -o grader grader.c Memory2.c -lm
```

- C++ 语言

```
1 | g++ -std=c++11 -O2 -o grader grader.cpp Memory2.cpp
```

如果编译成功，则会生成一个名为 `grader` 的可执行文件。

请注意，实际判题过程和样例的判题过程不同。样例判题程序作为单进程执行。这个程序需要从标准输入中读入，并输出到标准输出中。

Input

第一行三个整数 T, N, K ，由一个空格隔开。分别表示子任务编号为 T ，有 $2N$ 张牌，和 JOI 君最多问答 K 次。

第二行 N 个整数 $P_0, P_1, \dots, P_{N-1}$ ，表示 JOI 君对整数的记忆力。

第三行 $2N$ 个整数 $A_0, A_1, \dots, A_{2N-1}$ ，表示从左到右纸牌上写的整数。

Output

如果样例交互程序正常退出，它会向标准输出输出一行如下内容：

- 如果答案正确，输出 `Accepted`。
- 如果不正确，输出错误的类型，如 `Wrong answer [2]`。

Sample 1

Input	Output
1 3 10000 0 1 2 1 0 2 0 1 2	

Hint

样例交互

样例交互过程如下

函数调用	返回值
<code>Flip(0, 2)</code>	1

函数调用	返回值
Flip(0, 4)	1
Flip(1, 2)	0
Answer(0, 4, 1)	
Flip(1, 3)	0
Flip(5, 2)	2
Flip(4, 5)	1
Answer(1, 3, 0)	
Answer(5, 2, 2)	0

数据范围

所有输入数据满足以下条件：

- $1 \leq N \leq 50$ 。
- $0 \leq P_i \leq N - 1$ ($0 \leq i \leq N - 1$) 。
- $P_i \neq P_j$ ($0 \leq i < j \leq N - 1$) 。
- $0 \leq A_i \leq N - 1$ ($0 \leq i \leq 2N - 1$) 。
- 对于任意 x ($0 \leq x \leq N - 1$)，恰好存在两个 i ($0 \leq i \leq 2N - 1$) 满足 $A_i = x$ 。

子任务

子任务 1 [10 分]

满足以下条件：

- $T = 1$ 。
- $K = 10\,000$ 。
- $P_i = i$ ($0 \leq i \leq N - 1$) 。

子任务 2 [50 分]

满足以下条件：

- $T = 2$ 。
- $K = 400$ 。
- $P_i = i$ ($0 \leq i \leq N - 1$) 。

子任务 3 [40 分]

满足以下条件：

- $T = 3$ 。
- $K = 300$ 。

翻译由 Qwen3-235B 完成